SPORTS TOURNAMENT TEAM REGISTRATION AND MATCH RESULT SYSTEM

Complete Core Java Source Code

How to run: Save the code as **SportsTournamentSystem.java**. Open the VS Code terminal in that folder, then run `javac SportsTournamentSystem.java` and `java SportsTournamentSystem`.

```
import java.util.ArrayList;
import java.util.Scanner;

class Team {
    String teamName;
    String players;
    String category;

    Team(String teamName, String players, String category) {
        this.teamName = teamName;
        this.players = players;
        this.category = category;
    }
}

class Match {
    String team1;
    String team2;
    int score1;
    int score2;
    boolean resultEntered = false;

    Match(String team1, String team2) {
        this.team1 = team1;
        this.team2 = team2;
    }

    void enterResult(int score1, int score2) {
        this.score1 = score1;
        this.score2 = score2;
        resultEntered = true;
    }

    void displayResult() {
        System.out.println("\n--- MATCH RESULT ---");
        System.out.println("Match: " + team1 + " vs " + team2);

        if (!resultEntered) {
            System.out.println("Result: Not entered yet");
            return;
        }

        System.out.println("Score: " + score1 + " - " + score2);

        if (score1 > score2) {
            System.out.println("Winner: " + team1);
        } else if (score2 > score1) {
            System.out.println("Winner: " + team2);
        } else {
            System.out.println("Result: Draw");
        }
    }
}

public class SportsTournamentSystem {

    static Scanner sc = new Scanner(System.in);
    static ArrayList<Team> teams = new ArrayList<>();
    static ArrayList<Match> matches = new ArrayList<>();

    static void registerTeam() {
        System.out.print("Enter team name: ");
        String name = sc.nextLine().trim();

        if (name.isEmpty()) {
            System.out.println("Team name cannot be empty.");
            return;
        }

        for (Team t : teams) {
            if (t.teamName.equalsIgnoreCase(name)) {
                System.out.println("Team already registered!");
                return;
            }
```

```
    }

    System.out.print("Enter player names: ");
    String players = sc.nextLine();

    System.out.print("Enter sports category: ");
    String category = sc.nextLine();

    teams.add(new Team(name, players, category));

    System.out.println("Team registered successfully!");
}

static void viewTeams() {
    System.out.println("\n--- REGISTERED TEAMS ---");

    if (teams.isEmpty()) {
        System.out.println("No teams registered.");
        return;
    }

    for (int i = 0; i < teams.size(); i++) {
        Team t = teams.get(i);

        System.out.println("\nTeam Number: " + (i + 1));
        System.out.println("Team Name: " + t.teamName);
        System.out.println("Players: " + t.players);
        System.out.println("Category: " + t.category);
    }
}

static void scheduleMatch() {
    if (teams.size() < 2) {
        System.out.println("Register at least two teams first.");
        return;
    }

    viewTeams();

    System.out.print("\nEnter first team number: ");
    int first = readNumber();

    System.out.print("Enter second team number: ");
    int second = readNumber();

    if (first < 1 || first > teams.size()
            || second < 1 || second > teams.size()) {
        System.out.println("Invalid team number.");
        return;
    }

    if (first == second) {
        System.out.println("Choose two different teams.");
        return;
    }

    String team1 = teams.get(first - 1).teamName;
    String team2 = teams.get(second - 1).teamName;

    matches.add(new Match(team1, team2));

    System.out.println("Match scheduled successfully!");
    System.out.println(team1 + " vs " + team2);
}

static void enterResult() {
    if (matches.isEmpty()) {
        System.out.println("No matches scheduled.");
        return;
    }

    viewMatches();

    System.out.print("\nEnter match number: ");
    int number = readNumber();

    if (number < 1 || number > matches.size()) {
        System.out.println("Invalid match number.");
        return;
    }

    Match m = matches.get(number - 1);

    System.out.print("Enter score for " + m.team1 + ": ");
    int score1 = readNumber();

    System.out.print("Enter score for " + m.team2 + ": ");
```

```
    int score2 = readNumber();

    if (score1 < 0 || score2 < 0) {
        System.out.println("Scores cannot be negative.");
        return;
    }

    m.enterResult(score1, score2);

    System.out.println("Match result saved successfully!");
    m.displayResult();
}

static void viewMatches() {
    System.out.println("\n--- MATCH SCHEDULE ---");

    if (matches.isEmpty()) {
        System.out.println("No matches scheduled.");
        return;
    }

    for (int i = 0; i < matches.size(); i++) {
        Match m = matches.get(i);

        System.out.println((i + 1) + ". "
                + m.team1 + " vs " + m.team2);
    }
}

static void viewResults() {
    System.out.println("\n--- ALL MATCH RESULTS ---");

    if (matches.isEmpty()) {
        System.out.println("No matches available.");
        return;
    }

    for (Match m : matches) {
        m.displayResult();
    }
}

static int readNumber() {
    while (true) {
        try {
            int number = Integer.parseInt(sc.nextLine().trim());
            return number;
        } catch (NumberFormatException e) {
            System.out.print("Enter a valid whole number: ");
        }
    }
}

public static void main(String[] args) {

    int choice;

    do {
        System.out.println("\n================================");
        System.out.println(" SPORTS TOURNAMENT SYSTEM");
        System.out.println("================================");
        System.out.println("1. Register Team");
        System.out.println("2. View Registered Teams");
        System.out.println("3. Schedule Match");
        System.out.println("4. Enter Match Result");
        System.out.println("5. View Match Schedule");
        System.out.println("6. View All Results");
        System.out.println("7. Exit");
        System.out.println("================================");

        System.out.print("Enter your choice: ");
        choice = readNumber();

        switch (choice) {
            case 1:
                registerTeam();
                break;

            case 2:
                viewTeams();
                break;

            case 3:
                scheduleMatch();
                break;

            case 4:
```

```
                    enterResult();
                    break;

                case 5:
                    viewMatches();
                    break;

                case 6:
                    viewResults();
                    break;

                case 7:
                    System.out.println("Thank you for using the system!");
                    break;

                default:
                    System.out.println("Invalid choice. Try again.");
            }

        } while (choice != 7);

        sc.close();
    }
}
```